FCC-ue and Muon Collider: $HH \to 4b$ and $\nu\bar{\nu}4b$ Rates

Preliminary LO parton-level cross sections; $P_{\mu^+} = +80\%$

Beam-energy configurations

Collider	Incoming beams	$\sqrt{s}$
FCC-ue (μ^+e^-)	$E_{\mu^+} = 2.0\,\text{TeV}$, $E_{e^-} = 120\,\text{GeV}$	979.80 GeV
Muon collider ($\mu^+\mu^-$)	$E_{\mu^+} = E_{\mu^-} = 500\,\text{GeV}$	1.00 TeV

Fiducial selection

- All four final-state $b/\bar{b}$ partons satisfy $|\eta_b| < 5$.
- Form two *disjoint*, opposite-charge $b\bar{b}$ pairs. Accept the event if either distinct pairing has *both* masses in $110 \le m_{b\bar{b}}^{(1,2)} \le 130\,\text{GeV}$; count each event once.
- **No minimum p_T cut** is applied (in particular, the optional 5 GeV threshold is *not* used).
- These are parton-level estimates (no showering, jet clustering, tagging or detector simulation).

Cross sections before and after cuts

All entries in fb. Uncertainties shown after cuts are approximate Monte Carlo errors only.

Collider / beams	Final state	Before cuts	After cuts $\pm\delta_{\text{MC}}$
FCC-ue μ^+e^-	$\nu_e\bar{\nu}_\mu HH \to \nu_e\bar{\nu}_\mu\, 4b$	0.08898	**0.08816 ± 0.00015**
	$\nu_e\bar{\nu}_\mu\, 4b$	1.56028	**0.09195 ± 0.00213**
Muon collider $\mu^+\mu^-$	$\sum_{\ell=e,\mu,\tau} \nu_\ell\bar{\nu}_\ell\, HH(\to 4b)$	0.11593	**0.11582 ± 0.00017**
	$\sum_{\ell=e,\mu,\tau} \nu_\ell\bar{\nu}_\ell\, 4b$	1.72490	**0.12472 ± 0.00240**